



BULLETS · NUTRITION

Which health benefits of the Mediterranean diet are actually supported?

REFERENCES	TOKENS	VERIFICATION	COMPILED
97 verified	≈17.5k	Crossref	27 Aug 2026

SUMMARY

The clearest benefit is cardiovascular: supported Mediterranean-diet programmes reduce major events in people at elevated risk, usually by a modest absolute amount. Diabetes prevention is probable, and higher adherence is consistently associated with longer survival. Weight, liver fat, and metabolic markers improve modestly; cancer, dementia, and depression findings are promising but rely much more on observational studies, small trials, or indirect outcomes.

01 The intervention is a family of plant-forward patterns, not a fixed menu

- The recurring core is vegetables, fruit, legumes, nuts, minimally refined cereals, olive oil, and fish, with less red or processed meat and sweets. Dairy, alcohol, quantities, and scoring vary; adherence scores can reward different foods or population medians. [Davis et al. 2015](#), [Bach-Faig et al. 2011](#), [Bach et al. 2006](#), [Hutchins-Wiese et al. 2022](#)

- The 14-item Mediterranean Diet Adherence Screener (MEDAS) rapidly measures reported behaviour, not biological dose. [Schröder et al. 2011](#)
- The Seven Countries Study supplied the observational hypothesis that later trials tested; it was not causal proof. [Menotti & Puddu 2015](#)
- Review quality varies: among 24 cardiovascular syntheses, complete compliance with adapted methodological criteria ranged from 8% to 75%. [Huedo-Medina et al. 2016](#)
- The recurring core, variable definitions, and unsupported shortcuts are separated in [Figure 1](#).

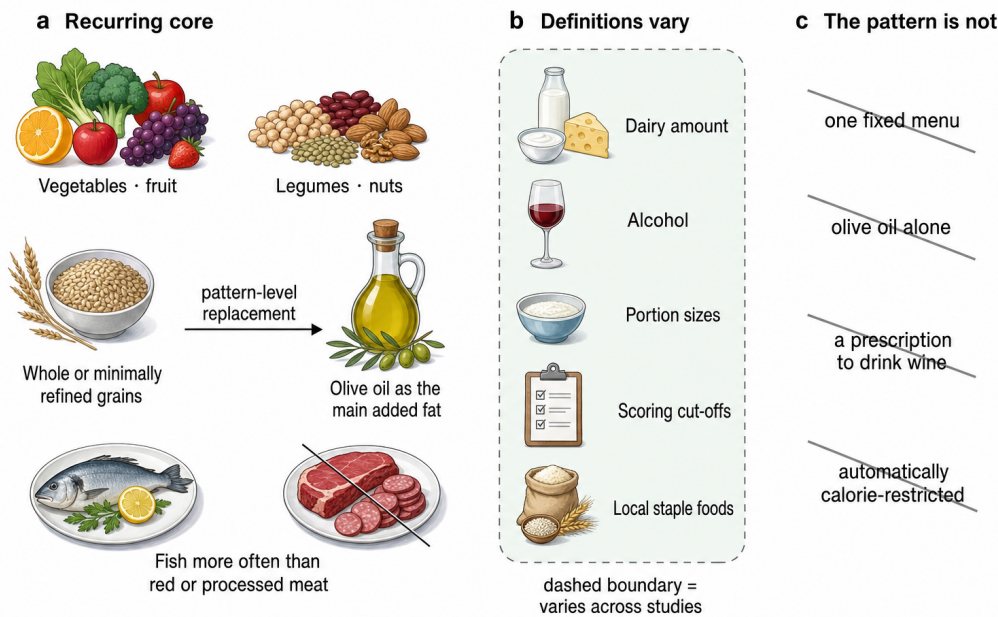


Figure 1. A Mediterranean diet is a pattern, not one fixed menu.

- **Shows:** The recurring food-group pattern, the elements that vary across studies, and four interpretations that the intervention evidence does not support.
- **Evidence boundary:** The replacement arrow is pattern-level synthesis, not a serving prescription or proof that one component causes the outcomes.
- **Sources:** [Davis et al. 2015](#), [Bach-Faig et al. 2011](#), [Hutchins-Wiese et al. 2022](#), [Keshani et al. 2024](#)

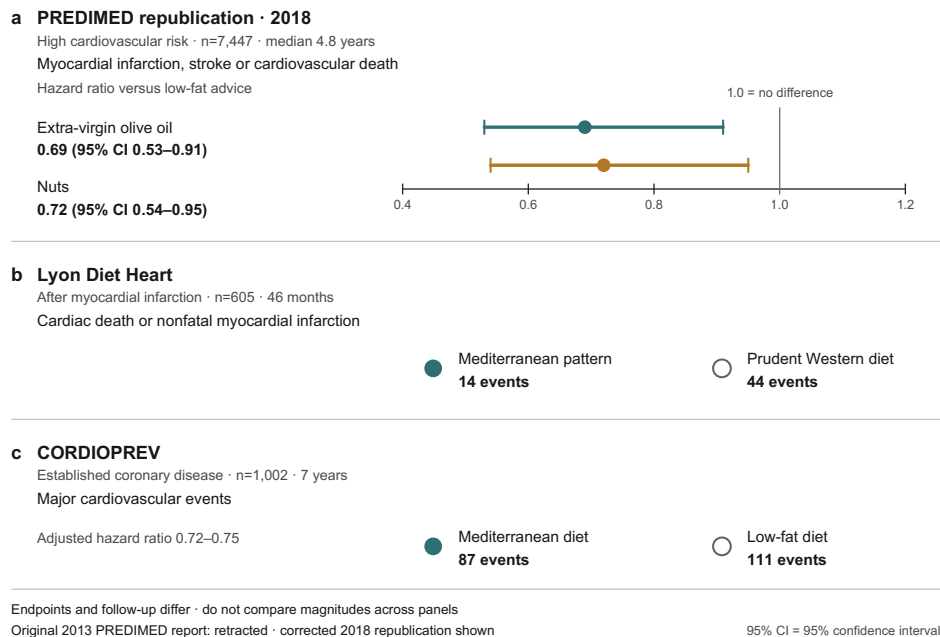
**02 Corrected PREDIMED and two secondary-prevention trials support fewer cardiovascular events**

- The original 2013 PREDIMED report is retracted and is not evidence here. After a statistical audit exposed unusual baseline distributions, investigators documented household enrolment and site-level allocation departures and issued a corrected 2018 republication. [Carlisle 2017](#), [Estruch et al. 2018](#)
- In the corrected analysis (7,447 high-risk adults; median 4.8 y), the primary composite occurred less often with extra-virgin olive oil (hazard ratio (HR) 0.69, 95% confidence interval (CI) 0.53–0.91) and nuts (0.72, 0.54–0.95) than with low-fat advice. Excluding 1,588 participants with known or suspected allocation departures produced essentially the same estimates. [Estruch et al. 2018](#)

- In the Lyon Diet Heart Study after myocardial infarction (n=605), cardiac death or nonfatal infarction occurred in 14 Mediterranean-pattern participants and 44 controls over 46 months; the magnitude is unusually large and has not been reproduced. [de Lorgeril et al. 1999](#)
- CORDIOPREV randomized 1,002 patients with coronary disease for 7 y: major events occurred in 87 Mediterranean-diet and 111 low-fat participants, with adjusted hazard ratios around 0.72–0.75. [Delgado-Lista et al. 2022](#)

TRIAL	POPULATION	COMPARATOR	MAIN RESULT	EVIDENCE KEY
PREDIMED republication	High cardiovascular risk; n=7,447	Low-fat advice	Major events: HR 0.69 with olive oil; 0.72 with nuts	Estruch et al. 2018
Lyon Diet Heart	Recent myocardial infarction; n=605	Prudent Western diet	Cardiac death/nonfatal infarction: 14 vs 44	de Lorgeril et al. 1999
CORDIOPREV	Established coronary disease; n=1,002	Low-fat diet	87 vs 111 major events; HR about 0.72–0.75	Delgado-Lista et al. 2022

- The exact trial results—and the reason their magnitudes should not be pooled visually—are shown in [Figure 2](#).

**Figure 2. Three endpoint trials support cardiovascular benefit.**

- **Shows:** Corrected PREDIMED hazard ratios and confidence intervals, plus the reported Lyon Diet Heart and CORDIOPREV event counts.
- **Evidence boundary:** Populations, comparators, endpoints, and follow-up differ; the 2013 PREDIMED report is retracted, and only the corrected 2018 republication is plotted.
- **Sources:** [Estruch et al. 2018](#), [de Lorgeril et al. 1999](#), [Delgado-Lista et al. 2022](#), [Karam et al. 2023](#)



03 Evidence syntheses confirm benefit over minimal support, not clear superiority over every healthy diet

- A network meta-analysis of 40 trials (35,548 participants at increased risk) found Mediterranean programmes superior to minimal intervention for all-cause mortality (odds ratio 0.72), cardiovascular mortality (0.55), stroke (0.65), and nonfatal infarction (0.48); at intermediate baseline risk this meant about 17 fewer deaths and 17 fewer infarctions per 1,000 over 5 y. [Karam et al. 2023](#)
- The same analysis found no convincing mortality or infarction advantage over low-fat programmes, which also outperformed minimal advice. [Karam et al. 2023](#)
- Earlier randomized-trial pooling found fewer major vascular events (risk ratio 0.63, 95% CI 0.53–0.75) but no all-cause mortality reduction (1.00, 0.86–1.15); excluding one unreliable trial weakened several secondary results. [Liyanage et al. 2016](#)
- Cochrane included 30 trials and 12,461 participants but judged most comparisons low or moderate certainty because interventions, comparators, reporting, and event counts differed. [Rees et al. 2019](#)
- Independent reviews reproduce lower cardiovascular event rates but stress that primary-prevention evidence leans heavily on PREDIMED and programmes often include counselling, free foods, or exercise. [Bloomfield et al. 2016](#), [Sebastian et al. 2024](#)
- The 2026 Italian-guideline syntheses again found lower primary and recurrent cardiovascular risk, but combined randomized and cohort evidence and reported certainty ranging from low to high by outcome. [Volpe et al. 2026a](#), [Volpe et al. 2026b](#)
- Compared with low-fat diets, Mediterranean patterns yield small favourable changes in weight, blood pressure, glucose, and lipids; between-diet differences are much smaller than comparisons with usual diet. [Nordmann et al. 2011](#)

04 Longer survival is a consistent association, but not a randomized mortality result

- In the foundational Greek cohort (22,043 adults), each 2-point adherence increase was associated with 25% lower total mortality; in 74,607 older Europeans the corresponding association was 8%, illustrating how estimates shrink across settings. [Trichopoulou et al. 2003](#), [Trichopoulou et al. 2005](#)
- Early pooling of 12 cohorts (1.57 million participants) estimated all-cause mortality risk ratio 0.91 per 2 adherence points; later meta-analyses reached similar 9%–10% reductions per 2 points. [Sofi et al. 2008](#), [Eleftheriou et al. 2018](#), [Soltani et al. 2019](#)
- The newest broad synthesis included 54 cohorts, 1.83 million participants, and 346,034 deaths: risk ratio 0.96 (95% CI 0.95–0.97) per 1-point increase, with moderate certainty. [Nucci et al. 2026](#)

- Two large US health-professional cohorts linked the top adherence category to total mortality HR 0.82 and cardiovascular disease HR 0.83; these are carefully adjusted but remain observational. [Shan et al. 2023](#), [Shan et al. 2020](#)
- Among 25,315 US women followed for up to 25 y, the highest adherence group had mortality HR 0.77; metabolic, inflammatory, lipoprotein, insulin-resistance, and body-mass pathways explained only part of the association. [Ahmad et al. 2024](#)
- Associations also appear in people with diabetes (mortality HR 0.63 per 2 points), yet healthier adherers differ in activity, smoking, education, and care access. [Bonaccio et al. 2016](#)
- Biomarker-based adherence predicted mortality better than repeated food questionnaires in an older Italian cohort, but imperfect specificity could not eliminate confounding. [Hidalgo-Liberona et al. 2021](#)

05 Weight and metabolic risk factors improve, mostly by small amounts

- A 50-study synthesis associated the pattern with lower metabolic-syndrome risk and pooled changes of roughly –2.4 mm Hg systolic pressure, –6 mg/dL triglycerides, –4 mg/dL glucose, and less than 1 cm waist circumference. [Kastorini et al. 2011](#)
- Across 57 controlled trials (36,983 participants), direction favoured the Mediterranean diet for 18 of 28 metabolic measures, but comparator and intervention heterogeneity was often high. [Papadaki et al. 2020](#)
- In DIRECT (n=322), 2-y weight loss was 4.4 kg with Mediterranean, 2.9 kg with low-fat, and 4.7 kg with low-carbohydrate diets; thus the pattern beat low fat but not low carbohydrate. [Shai et al. 2008](#)
- Across 121 trials, most named diets produced modest 6-mo losses that faded by 12 mo; Mediterranean programmes retained some cardiovascular risk-factor improvement but no unique weight-loss advantage. [Ge et al. 2020](#)
- PREDIMED-Plus combined a 30% energy reduction, activity, and intensive behavioural support. In a 1,521-person imaging substudy, it reduced total and visceral fat and slowed relative lean-mass loss over 3 y, but diet alone cannot receive the credit. [Konieczna et al. 2023](#)
- A polyphenol-enriched green variant reduced visceral fat by 14.1% versus 6.0% with standard Mediterranean and 4.2% with healthy guidance; 88% of participants were men in a monitored workplace. [Zelicha et al. 2022](#)
- A 2024 review of olive-oil-enriched Mediterranean trials found no consistent cardiometabolic benefit beyond a very small triglyceride reduction, countering claims that extra oil alone reproduces the whole pattern. [Keshani et al. 2024](#)



- In pre-existing metabolic disease, 69-study pooling found modest improvements in body mass index, waist, low-density lipoprotein (LDL) cholesterol, triglycerides, glucose, insulin resistance, and C-reactive protein (CRP), while glycated haemoglobin (HbA1c) and lean-mass findings remained inconsistent. [Muscogiuri et al. 2026](#)

06 Type 2 diabetes has unusually coherent trial, cohort, and biomarker evidence

- In the diabetes-free PREDIMED subgroup, Mediterranean diets reduced incident diabetes relative to low-fat advice; annual adherence analysis gave HR 0.80 (95% CI 0.70–0.92) per 2 MEDAS points and 0.46 (0.25–0.83) for highest versus lowest adherence. [Salas-Salvadó et al. 2014](#), [Martínez-González et al. 2023](#)
- PREDIMED-Plus later found that adding energy restriction and activity reduced 6-y diabetes incidence by 31% relative to an unrestricted Mediterranean diet (9.5% vs 12.0%), again testing a multicomponent programme rather than diet alone. [Ruiz-Canela et al. 2025](#)
- A Mediterranean biomarker score trained in a feeding trial predicted diabetes HR 0.71 (0.65–0.77) per standard deviation in EPIC-InterAct (22,202 participants), compared with 0.90 for self-report; specificity and residual confounding remain possible. [Sobiecki et al. 2023](#)
- In 112,493 UK Biobank participants, the top Mediterranean-lifestyle quartile had diabetes HR 0.70, but the exposure deliberately included physical activity, rest, and social habits. [Maroto-Rodriguez et al. 2023](#)
- Systematic reviews of prospective studies estimate 19%–23% lower incidence with higher adherence. In established diabetes, randomized evidence shows modest improvements rather than remission by diet alone. [Schwingshackl et al. 2015](#), [Esposito et al. 2015](#)
- A 2025 synthesis of 11 randomized controlled trials found HbA1c –0.31 percentage points, fasting glucose –0.85 mmol/L, and body mass index –0.83 kg/m² versus control diets. [Wu et al. 2025](#)

CARDIOMETABOLIC CLAIM	BEST ESTIMATE	INTERPRETATION	EVIDENCE KEY
Incident diabetes, PREDIMED-Plus	9.5% vs 12.0% over 6 y	Additional benefit from energy restriction + activity	Ruiz-Canela et al. 2025
HbA1c in established diabetes	–0.31 percentage points	Small but clinically useful adjunct	Wu et al. 2025
Weight in DIRECT	–4.4 kg at 2 y	Better than low fat; similar to low carbohydrate	Shai et al. 2008
Olive-oil-enriched trials	Mostly null	Oil supplementation is not the whole pattern	Keshani et al. 2024

07 Cancer associations are broad, but causal evidence is thin

- An umbrella review rated support strongest for overall cancer incidence and mortality, while most site-specific findings were only suggestive or weak. [Dinu et al. 2018](#)
- The largest updated synthesis (117 studies; 3.20 million participants) associated high adherence with cancer mortality risk ratio 0.87, colorectal cancer 0.83, and breast cancer 0.94; most site-specific certainty was low or very low. [Morze et al. 2021](#)
- Cancer meta-analyses converge on inverse observational associations, but varying scores, retrospective data, and lifestyle confounding limit causal interpretation. [Schwingshackl et al. 2017](#)
- Dose-response pooling supports lower colorectal risk, while a separate postmenopausal breast-cancer cohort/meta-analysis found protection concentrated in estrogen-receptor-negative disease; neither is randomized. [Zhong et al. 2020](#), [van den Brandt & Schulpen 2017](#)
- PREDIMED reported fewer invasive breast cancers in the olive-oil arm (hazard ratio 0.32, 95% CI 0.13–0.79), but this secondary analysis contained only 35 cases. [Toledo et al. 2015](#)
- Olive-oil intake itself was associated with cardiovascular, diabetes, and mortality benefits but not cancer (risk ratio 0.94, 0.86–1.03), weakening single-ingredient explanations. [Martínez-González et al. 2022](#)
- A breast-cancer umbrella review likewise found consistent direction but predominantly observational evidence, leaving prevention trials as the missing test. [González-Palacios Torres et al. 2023](#)

08 Dementia cohorts are positive; the largest dedicated cognitive trial was null

- Meta-analyses of observational studies associate higher adherence with cognitive impairment HR 0.82, dementia 0.89, and Alzheimer disease 0.70, but report heterogeneity and possible publication bias. [Fekete et al. 2025](#), [Nucci et al. 2024](#)
- UK Biobank followed 60,298 adults for 9.1 y: high adherence predicted dementia HR 0.77 and absolute risk 1.18% versus 1.73%, independent of measured genetic risk. [Shannon et al. 2023](#)
- Earlier synthesis found mild cognitive impairment risk ratio 0.75 and Alzheimer disease 0.71 in cohorts; two small trials yielded episodic-memory standardized mean difference 0.20. [Fu et al. 2022](#), [Singh et al. 2014](#)
- A PREDIMED substudy reported better global cognition after 6.5 y, but it was small and inherited the parent trial's allocation concerns. [Martínez-Lapiscina et al. 2013](#)



- In contrast, the 3-y MIND trial (604 older adults; calorie-restricted intervention and control) found a global-cognition difference of 0.035 standard deviations (95% CI –0.022 to 0.092) and no meaningful [magnetic resonance imaging \(MRI\)](#) difference. [Barnes et al. 2023](#)
- A MIND-diet meta-analysis remains favourable because cohort weights dominate, while a 2024 PREVENT analysis found no association with cognition or neuroimaging. [Chen et al. 2023](#), [Gregory et al. 2024](#)

09 Depression trials point in the same direction, but effect size is unstable

- Prospective dietary-pattern studies associate high adherence with roughly one-third lower incident depression, but reverse causation and social-pattern confounding remain plausible. [Lassale et al. 2019](#), [Shafiei et al. 2019](#)
- SMILES randomized 67 adults for 12 wk: the diet arm improved [Montgomery-Åsberg Depression Rating Scale](#) scores 7.1 points more than social support; remission was 32% versus 8%. [Jacka et al. 2017](#)
- HELFIMED (n=152) combined Mediterranean advice with fish-oil supplementation and group delivery, producing greater symptom improvement than social groups; the design cannot isolate the diet. [Parletta et al. 2019](#)
- AMMEND randomized 72 young men and found larger symptom reductions over 12 wk, extending the signal to another small population-specific trial. [Bayes et al. 2022](#)
- Five-trial pooling (1,507 participants) estimated symptom standardized mean difference –0.53, but heterogeneity was 87%, the prediction interval crossed zero, and certainty was low. [Bizzozero-Peroni et al. 2025](#)
- A 104-study mental-health review found broadly favourable associations, but only 16 studies were interventional. [Kabthlymer et al. 2026](#)

10 Liver fat falls with calorie-restricted variants, without clear superiority to low-fat diets

- An ad libitum Mediterranean diet and a low-fat diet both reduced liver fat in a 12-wk randomized trial, with no decisive between-diet advantage. [Properzi et al. 2018](#)
- In DIRECT, hepatic-fat reduction statistically mediated part of the Mediterranean diet's cardiometabolic advantage over low fat, but mediation does not identify which foods caused it. [Gepner et al. 2019](#)
- DIRECT-PLUS (n=294) found relative liver-fat reductions of 38.9% with green Mediterranean, 19.6% with standard Mediterranean, and 12.2% with healthy guidance at 18 mo; all groups received activity support and the enriched arm included green tea, walnuts, and Mankai. [Yaskolka Meir et al. 2021](#)
- Earlier randomized-trial pooling suggested improvements in steatosis and insulin resistance, whereas enzyme effects were small and inconsistent. [Kawaguchi et al. 2021](#), [Sangouni et al. 2022](#)

- A 2024 head-to-head meta-analysis concluded Mediterranean and low-fat diets were similarly effective for short-term liver fat and enzymes, a direct contradiction to claims of unique superiority. [Xiong et al. 2024](#)
- An Asian-adapted trial supports transferability for fatty-liver management, but changes the tested pattern and leaves long-term outcomes unknown. [Chooi et al. 2024](#)

11 Measured biology is plausible and responsive, but does not substitute for clinical outcomes

- Trial meta-analysis finds small improvements in endothelial function and inflammatory markers, though effects differ by assay, population, and comparator. [Schwingshackl & Hoffmann 2014](#), [Koelman et al. 2022](#)
- A 2026 synthesis of 33 trials (3,476 participants) found reductions in high-sensitivity CRP, interleukin-6, and interleukin-17, but not conventional CRP, interleukin-10, tumour-necrosis factor, or total antioxidant capacity. [Keshani et al. 2026](#)
- Mechanistic reviews therefore support several converging pathways—fat quality, polyphenols, fibre fermentation, inflammation, endothelial function, insulin sensitivity—without identifying one necessary ingredient. [Itsiopoulos et al. 2022](#)
- In an 8-wk isocaloric trial (n=82), the diet lowered total cholesterol and altered bile acids, the microbiome, and metabolome without changing body weight; glucose, insulin, and CRP were null. [Meslier et al. 2020](#)
- A 1-y older-adult intervention (microbiome subset n=612) shifted microbial taxa and functions that covaried with lower frailty and inflammation, while a separate obesity trial showed that two healthy diets altered gut communities and insulin sensitivity. [Ghosh et al. 2020](#), [Haro et al. 2016](#)
- Plasma metabolomic profiles partly tracked adherence and cardiovascular risk, providing objective exposure measures rather than proof that the metabolites mediate events. [Li et al. 2020](#)
- Baseline bile acids and microbes may modify lipid and adiposity responses; PREDIMED-Plus also links diet-induced microbiome and metabolome shifts with risk factors. These are secondary analyses, not independent endpoint trials. [Gao et al. 2024](#), [García-Gavilán et al. 2024](#)

12 Translation depends on culture, cost, support, and what replaces meat or alcohol

- Outside the Mediterranean, food availability, cooking, resources, and adherence scores change; cultural adaptation is part of the intervention. [Hoffman & Gerber 2013](#), [Woodside et al. 2022](#)

- Score agreement is limited even within one study population, so an effect per point is not automatically portable between instruments or countries. [Olmedo-Requena et al. 2019](#)
- MedEx-UK demonstrated that a theory-informed programme could improve adherence and remain acceptable in older adults at dementia risk, but the intervention supplied behavioural maintenance and activity support. [Jennings et al. 2024](#)
- Price, access, cooking skills, family norms, and knowledge impede adherence; free foods, counselling, and frequent contact make trials more supportive than routine life. [Biggi et al. 2024](#)
- Economic modelling in secondary prevention suggests Mediterranean diet plus activity can be cost-effective, but estimates inherit assumptions about adherence and event reduction. [Bonekamp et al. 2024](#)
- A sustainability review (35 studies, mostly models) classified the pattern as environmentally favourable in 91% of comparisons, with results driven mainly by lower ruminant-meat intake; water footprints and local production can reverse particular comparisons. [Lorca-Camara et al. 2024](#), [Dernini et al. 2017](#)
- Traditional scores reward moderate alcohol, but modern interpretation should not encourage non-drinkers to start: cancer and addiction risk make wine an optional historical component, not a therapeutic ingredient. [Martínez-González 2024](#)
- Dietary-pattern mortality reviews still identify a much larger observational than randomized evidence base, so residual healthy-user bias remains a central limit even after extensive adjustment. [English et al. 2021](#)
- The intervention package, adherence conditions, and causal stopping point are traced in [Figure 3](#).

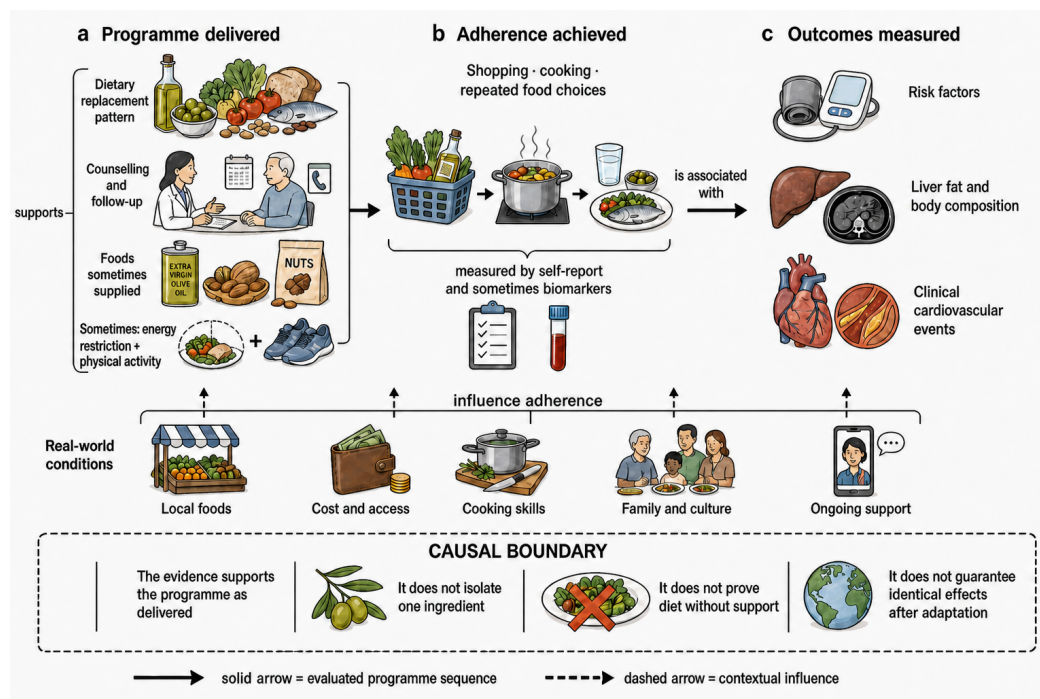


Figure 3. Benefits belong to the supported programme as delivered.

- **Shows:** Trials evaluate a replacement pattern with varying support; real-world conditions influence adherence before risk factors, imaging outcomes, or clinical events are measured.
- **Evidence boundary:** The pathway is programme logic, not a quantified mediation model; it does not isolate one ingredient, prove unsupported diet alone, or guarantee identical effects after adaptation.
- **Sources:** [Estruch et al. 2018](#), [Jennings et al. 2024](#), [Biggi et al. 2024](#), [Lorca-Camara et al. 2024](#), [Ruiz-Canela et al. 2025](#)

13 The evidence supports a hierarchy of claims, not a universal health halo

OUTCOME	BEST-SUPPORTED CONCLUSION	MAIN UNCERTAINTY
Major cardiovascular events	Probably reduced in higher-risk adults, especially versus minimal advice	Absolute benefit and superiority to other high-quality diets Karam et al. 2023
Type 2 diabetes	Probably prevented; glycaemia improves modestly in established disease	Added effects of calorie restriction, activity, and intensive support Ruiz-Canela et al. 2025
All-cause mortality	Consistently associated with 4% lower risk per adherence point	No mortality trial capable of confirming the cohort estimate Nucci et al. 2026
Weight and metabolic markers	Small, clinically useful changes	Not uniquely better than low-carbohydrate or low-fat diets Ge et al. 2020

OUTCOME	BEST-SUPPORTED CONCLUSION	MAIN UNCERTAINTY
Cancer	Lower incidence/mortality associations for several sites	Mostly observational; one small secondary randomized signal Morze et al. 2021
Cognition	Cohorts favourable; dedicated 3-y trial null	Duration, population risk, active comparator, residual confounding Barnes et al. 2023
Depression	Small trials favourable	Low certainty, high heterogeneity, weak blinding Bizzozero-Peroni et al. 2025
Fatty liver	Liver fat improves with energy-restricted variants	Similar short-term efficacy to low-fat diets; few hard outcomes Xiong et al. 2024

- The most defensible interpretation is selective: use the pattern as a supported cardiovascular and diabetes-prevention programme, not as proof that olive oil, wine, or any single food prevents every chronic disease.
- For outcomes beyond cardiometabolic disease, larger preregistered trials with active comparators, objective adherence measures, diverse populations, and clinical rather than surrogate endpoints would change confidence most.
- The uneven directness and certainty across outcomes are mapped in [Figure 4](#).

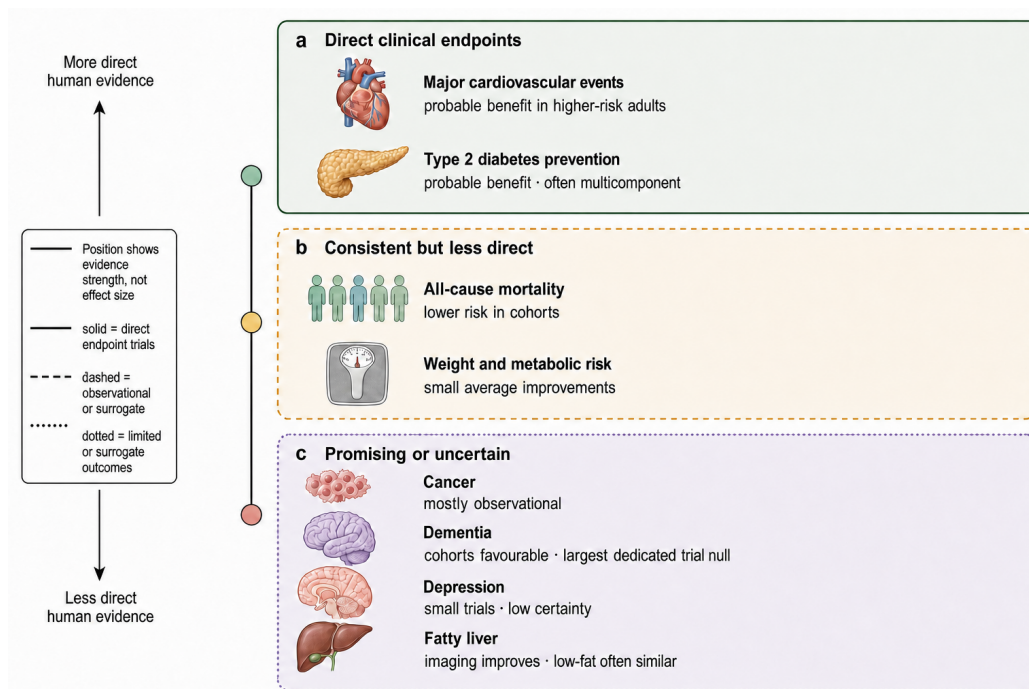


Figure 4. The evidence is strongest for cardiometabolic outcomes.

- **Shows:** Outcomes are arranged by human evidence directness and certainty, from clinical endpoint trials through consistent but less-direct evidence to limited or surrogate outcomes.
- **Evidence boundary:** This is a narrative-review synthesis, not a formal GRADE score; position, marker area, and colour do not encode effect size or individual benefit.
- **Sources:** [Karam et al. 2023](#), [Ruiz-Canela et al. 2025](#), [Nucci et al. 2026](#), [Morze et al. 2021](#), [Barnes et al. 2023](#)

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